

Felipe Cybis Pereira

DATA SCIENCE – NEUROSCIENCE ENGINEER – RESEARCH SOFTWARE ENGINEER

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Experience 🏠

Postdoctoral Researcher, neurosciences, functional ultrasound imaging

Paris, France

@ PHYSICS FOR MEDICINE PARIS, INSERM, ESPCI PARIS, PSL, CNRS 🌐

2025-2026

- Adviser: Sophie Pezet, PI, and Mickaël Tanter, PI.
- Focus: Data analysis for functional ultrasound imaging (fUSI) and Ultrasound Localization Microscopy (ULM).
- Techniques: **Data analysis for fUSI and ULM - Data Visualization - Python - General Linear Modeling - Skeletonization**

Google Summer of Code 2025

Campinas, Brazil

@ INVESALIUS 🌐

2025

- Mentors: Thiago F. Moraes, and Paulo Henrique Junqueira Amorim.
- Focus: Allow editing entire medical imaging volumes simultaneously via 3D projections 🌐
- Techniques: **3D visualization - Qt and VTK (Python) - Cython - Scientific open-source software development**

PhD, acoustics, neuroscience, spatial navigation and functional ultrasound imaging

Paris, France

@ PHYSICS FOR MEDICINE PARIS, INSERM, ESPCI PARIS, PSL, CNRS 🌐

2021-2025

- Adviser: Sophie Pezet, PI, and Mickaël Tanter, PI.
- Title: Revealing vascular patterns in the spatial navigation system with functional ultrasound imaging 🌐
- Recognition: PariSanté Campus PhD Award 2025 (Innovation & Industry) 🌐
- Techniques: **Functional ultrasound imaging - Python - General Linear Modeling - Brain decoding - Rodent handling, behavior and cognition - 3D design conception (CAD)**

Research Intern, neurobiology and ethology

Boston, USA

@ ROGULJA LAB, HARVARD MEDICAL SCHOOL, HARVARD UNIVERSITY 🌐

2019

- Adviser: Alexandra Vaccaro, PhD, and Dragana Rogulja, PI.
- Focus: Insights in sleep deprivation in *Drosophila melanogaster*.
- Techniques: ***Drosophila melanogaster* rearing - MATLAB - Immunostaining - Confocal microscopy - Survival assays**

Skills 🌐

- **Scientific Python**: Neuroimaging (**Nipy suite**, **BrainGlobe**), ML (**scikit-learn**, **PyTorch**), Visualization (**napari**, **VTK**, **PyVista**).
- **Packaging**: Publishing, API documentation and examples, unit testing, linting and formatting, pre-commit hooks.
- **Version control**: Advanced usage of **Git**, **GitHub** and **GitHub Actions**.
- **Prototyping**: Intermediate knowledge in **Arduino** and **Raspberry Pi**.
- Intermediate knowledge on **Rust** (and **Rust bindings for Python**), **Lua**, **core-utils** and **shell-scripting**, **JavaScript**.
- Intermediate knowledge on text editing softwares such as **LaTeX** and **Typst**.

Teaching 📖

Teacher assistant: Bootcamp - Introduction to Python

Paris, France

@ M1 AIRE, LEARNING PLANET INSTITUTE

2026

- From variables, functions and lists to basic numpy, matplotlib, pandas.

Teacher assistant: Methods for measuring and actuating neuronal activity

Saclay, France

@ M2 COMPUTATIONAL NEUROSCIENCES AND NEUROENGINEERING, NEUROPSI

2023-2026

- Principles on Ultrafast Ultrasound, functional ultrasound imaging and Ultrasound Localization Microscopy.

Teacher assistant: Practical work in physiology and neuroscience

Paris, France

@ ESPCI PARIS - PSL UNIVERSITY

2022-2024

- Neuroscience-focused practical work: (1) human EEG (2) pose-estimation and animal tracking (3) human sleep.

Education 🎓

Doctorate Degree in Acoustics (applied to Neurosciences)

Paris, France

@ UNIVERSITÉ PARIS SCIENCES ET LETTRES (PSL)

2021-2025

Master's Degree in Bioengineering and Neurosciences

Paris, France

@ BIOMEDICAL ENGINEERING MASTER

2020-2021

Engineering Degree (Biotechnology major)

Paris, France

@ ESPCI PARIS - PSL

2016-2019

Bachelor's Degree in Chemical Engineering

Florianópolis, Brazil

@ UNIVERSIDADE FEDERAL DE SANTA CATARINA (UFSC)

2014-2020

Hobbies 🧑

I like playing **Tennis** (I played a lot while kid/teen), programming and tweaking my own dotfiles.

Publications and preprints

1. Khallaf, M.A., Hart, D.W., Luo, W., Murad, F., **Cybis Pereira, F.**, Mendez-Aranda, D., Hagenah, N., Rossi, A., Bégay, V., Okrouhlík, J., et al. (2026). A Queen Odour Mediates Reproductive Suppression in a Eusocial Mammal. *Nature*, 656, 390–398. <https://doi.org/10.1038/s41586-026-10772-5>.
2. Le Meur-Diebolt, S., **Cybis Pereira, F.**, Mariani, J.-C., Kliewer, A., Farinha-Ferreira, M., Bertolo, A., Osmanski, B.-F., Lenkei, Z., and Deffieux, T. (2026). Robust Functional Ultrasound Imaging in the Awake and Behaving Brain: A Systematic Framework for Motion Artifact Removal. *Imaging neuroscience (Cambridge, Mass.)*, 4. <https://doi.org/10.1162/IMAG.a.1191>.
3. **Cybis Pereira, F.**, Castedo, S.H., Meur-Diebolt, S.L., Ialy-Radio, N., Bhattacharya, S., Ferrier, J., Osmanski, B.F., Cocco, S., Monasson, R., Pezet, S., et al. (2026). A Vascular Code for Speed in the Spatial Navigation System. *Cell Reports*, 45. <https://doi.org/10.1016/j.celrep.2025.116791>.
4. **Cybis Pereira, F.** (2025). Revealing Vascular Patterns of the Spatial Navigation System Using Functional Ultrasound Imaging. *Université Paris sciences et lettres (PSL University)*. Doctorat En Acoustique, PhD. <https://doi.org/10.70675/3815cfb5z4f06z42fezaf29z155a835e38b8>.
5. Zucker, N., Le Meur-Diebolt, S., **Cybis Pereira, F.**, Baranger, J., Hurvitz, I., Demené, C., Osmanski, B.-F., Ialy-Radio, N., Biran, V., Baud, O., et al. (2025). Physio-fUS: A Tissue-Motion Based Method for Heart and Breathing Rate Assessment in Neurofunctional Ultrasound Imaging. *eBioMedicine*, 112, 105581. <https://doi.org/10.1016/j.ebiom.2025.105581>.

International conferences

6. Le Meur-Diebolt, S., and **Cybis Pereira, F.** (2026). ConfUSlus: An Accessible and Scalable Toolkit for Functional Ultrasound Imaging Analysis. In *fUSbrain 2026*.
7. Huang, J., Maguelone, F., Vetere, G., Pezet, S., Travert-Jouanneau, Y., **Cybis Pereira, F.**, Melik-Parsadaniantz, S., Amar, L., Bourgeois Rambur, L., and Réaux-Le-Goazigo, A. (2024). Deciphering the Precise C-Fos Connectome of Ocular Pain in Mice. In *2024 ARVO Annual Meeting*, p. 2637.
8. **Pereira, F.C.**, Ialy-Radio, N., Bhattacharya, S., Nouhoum, M., Osmanski, B.-F., Pezet, S., and Tanter, M. (2024). Functional Ultrasound Tools for Automatic Atlas Registration and Chronic Neuroimaging on Naturally Behaving and Sleeping Rats. *Sleep Medicine*, 115, S409. <https://doi.org/10.1016/j.sleep.2023.11.1098>.
9. Bourgeois Rambur, L., Travert, Y., **Cybis Pereira, F.**, Huang, J., Baudouin, C., Melik-Parsadaniantz, S., Deffieux, T., Pezet, S., and Réaux-Le-Goazigo, A. (2023). Ultrafast Ultrasound Imaging of the Trigeminal Ganglion and Brain in Mice. In *2023 ARVO Annual Meeting (Investigative Ophthalmology & Visual Science)*, p. 3361.
10. **Cybis Pereira, F.**, Ialy-Radio, N., Bhattacharya, S., Osmanski, B.-F., Pezet, S., and Tanter, M. (2022). Chronic Functional Ultrasound Imaging Combined with Behaviour Tracking on Freely Moving Rats. In *Neuroscience 2022 (Society for Neuroscience)*.
11. **Cybis Pereira, F.**, Ialy-Radio, N., Bhattacharya, S., Osmanski, B.-F., Pezet, S., and Tanter, M. (2022). Chronic Functional Ultrasound Imaging Combined with Behaviour Tracking on Freely Moving Rats. In *fUSbrain 2022*.